

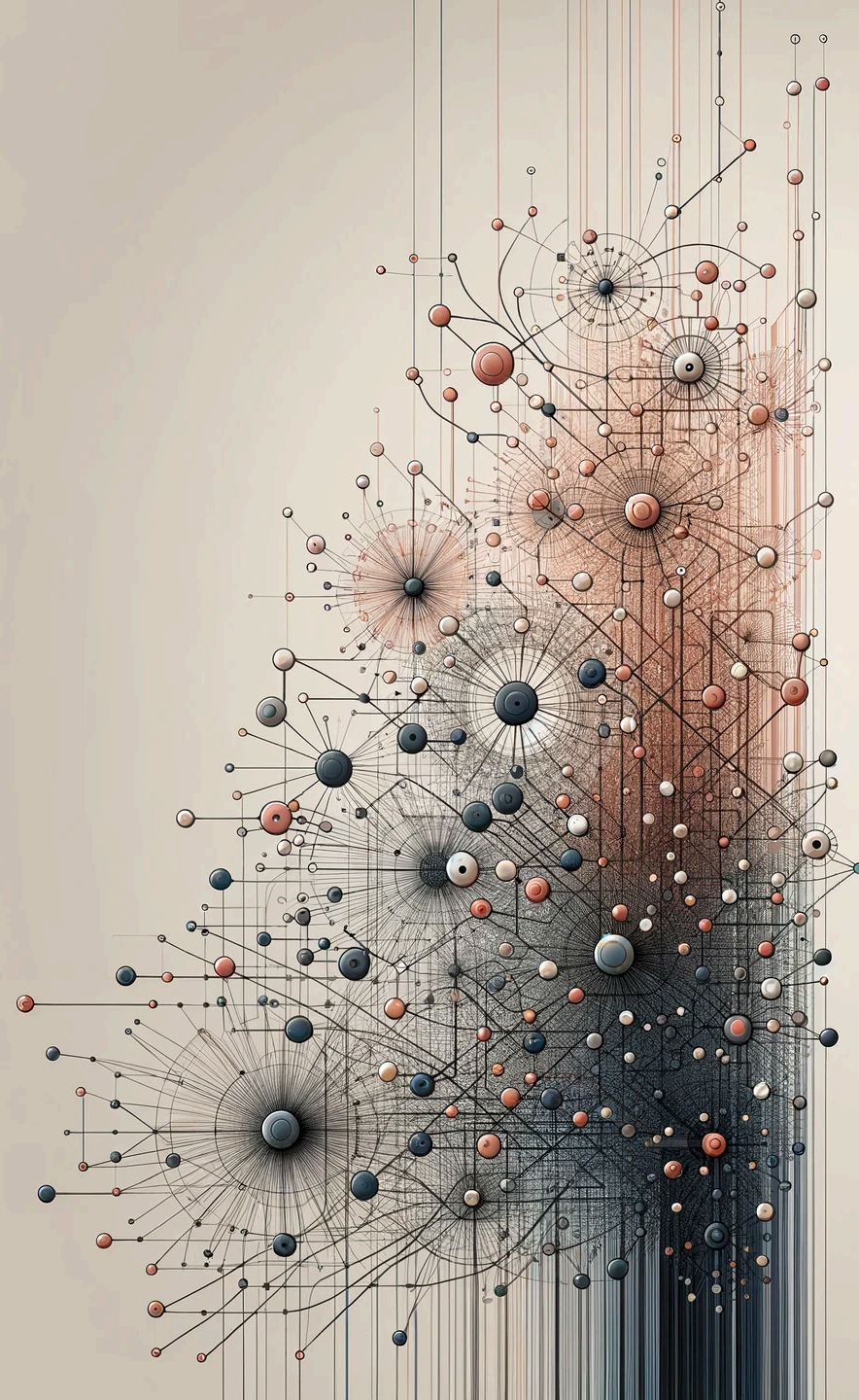


UNIVERSITÀ
DI TORINO

Analisi e Visualizzazione delle Reti Complesse

NS23 - Computational Epidemiology — Network Models

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Outline

1. **Why networks change the picture** — limits of homogeneous mixing; the network representation
2. **Three approaches** — homogeneous mean-field, heterogeneous mean-field, individual-based / spectral
3. **Comparing the approaches** — what each one buys, and what it costs
4. **Interventions on networks** — random, targeted, and the friendship-paradox trick

A short story — the superspreader

In 2003, a single physician in a Hong Kong hotel infected at least 16 other guests in a single evening — *seeding outbreaks of SARS in five countries within a week*.

That one person did not have a higher R_0 than anyone else *on average*. He had **far more contacts than average** — and the average is what mean-field models track.

This is the story NS22 cannot tell. NS23 fixes it.

What is special about networks?

Mean-field models from NS22 know exactly **two** things about your contact structure:

- the average number of contacts $\langle k \rangle$
- nothing else

What they cannot represent:

- That you might have **5 close friends** while your most popular classmate has **500**
- That your friends know **each other** (clustering)
- That you and your professor live in different **communities** of the contact graph
- That a single **bridge** between two communities can ignite an outbreak in a previously-untouched region

The price of mean-field's mathematical simplicity is that all of this is invisible. Networks let us put it back in.

What we will learn to compute

By the end of this lecture, you will be able to:

- **Predict the epidemic threshold** of a real (heterogeneous) network — and see why it can effectively *vanish* on scale-free networks
- **Compute** R_0 in three increasingly realistic ways — by average degree, by degree distribution, by full adjacency matrix
- **Design interventions** that beat random vaccination by orders of magnitude
- Apply the **friendship paradox** to find hubs without knowing the network

The disease model (SIR) stays the same. What changes is the contact structure on which it runs.

Beyond homogeneous mixing

The compartmental models in NS22 all assume **homogeneous mixing** — *"anyone can infect anyone with equal probability."* Convenient mathematically; wrong in reality.

What gets ignored:

- Heterogeneity in **contact patterns** (some people meet far more others than the average)
- Different **interaction types** (close family vs. casual passers-by)
- **Social structure** — communities, schools, workplaces, households
- **Geographic distribution**

Consequences of pretending these don't exist:

- **Overestimates** epidemic size in clustered populations (the disease keeps re-infecting the same group)
- **Underestimates** early spread in heterogeneous networks (hubs amplify the early phase)
- **Inaccurately predicts** the effects of targeted interventions (because targeting only matters if there is heterogeneity to target)
- **Cannot explain superspreading events at all**

Network representation in epidemiology

Why networks are the natural fix.

- Explicitly model **who can infect whom**
- Capture **heterogeneity** in contact patterns
- Enable **realistic modelling** of transmission dynamics

Formal setup.

- Network $G = (V, E)$ where:
 - V = set of nodes (individuals)
 - E = set of edges (contacts)
- Each node i has a discrete state: S , I , or R
- Infection spreads along network edges, not through some homogeneous "ether"

The disease model (SIR) is unchanged. What changes is *the contact structure on which it runs*.

From compartmental to network — three levels of detail

The big modelling question:

*How can we extend epidemic models to networks while balancing **complexity** and **accuracy**?*

The literature offers three classical answers, in order of increasing realism (and computational cost):

- **Homogeneous mean-field** — keep everything aggregate, just use the network's *average degree*
 - Preserves mathematical tractability while adding a basic network property
- **Heterogeneous mean-field** — group nodes by *degree class*
 - Critical for realistic modelling of hub effects in scale-free networks
- **Individual-based / spectral** — use the *full* adjacency matrix
 - Captures stochastic effects and enables targeted interventions

The next part walks through each in turn.

Approach 1 — Homogeneous Mean-Field

Main idea. Extend the classic SIR model to networks by replacing "everyone meets everyone" with "everyone meets $\langle k \rangle$ others on average."

Assumption. All individuals are statistically equivalent and have, on average, the same number of contacts.

- Simple and mathematically tractable
- Ignores differences in individual connectivity — only the *average* degree appears

This is the smallest possible step away from mean-field — barely a network model at all.

SIR equations — Homogeneous Mean-Field

$$\begin{aligned}\frac{dS(t)}{dt} &= -\beta \langle k \rangle \frac{S(t)I(t)}{N} \\ \frac{dI(t)}{dt} &= \beta \langle k \rangle \frac{S(t)I(t)}{N} - \gamma I(t) \\ \frac{dR(t)}{dt} &= \gamma I(t)\end{aligned}$$

Where:

- $S(t), I(t), R(t)$ — numbers of susceptible, infected, and recovered individuals at time t
- N — total population size
- $\langle k \rangle$ — average degree (average number of contacts per person)
- β — transmission rate per contact, γ — recovery rate

The only change from NS22's SIR is the multiplicative factor $\langle k \rangle$ on the infection rate.

Where the equations come from — step by step

1. **Contact structure.** Each person has $\langle k \rangle$ contacts on average.
2. **Probability of infectious contact.** The chance that a given contact is currently infected is $I(t)/N$.
3. **Expected infectious contacts.** Each susceptible has $\langle k \rangle \cdot I(t)/N$ infectious contacts.
4. **Force of infection.** Each infectious contact transmits at rate β .
5. **Total new infections.** Sum over the susceptible population: $\beta \langle k \rangle S(t) I(t)/N$.

That's the term that drives the dS/dt equation — and the new infections it produces are what flow into I .

Key quantities — Homogeneous Mean-Field

- **Basic reproduction number:** $R_0 = \frac{\beta \langle k \rangle}{\gamma}$
- **Critical transmission rate:** $\beta_c = \frac{\gamma}{\langle k \rangle}$
- **Early-stage growth exponent:** $\alpha = \beta \langle k \rangle - \gamma = \gamma(R_0 - 1)$

Compared to the no-network SIR ($R_0 = \beta/\gamma$), the network version simply scales by $\langle k \rangle$ — more contacts per person means a proportionally faster epidemic.

Interpretation and limits — Homogeneous Mean-Field

- R_0 . Increases linearly with average degree $\langle k \rangle$. More contacts per person $\rightarrow$ faster, larger outbreaks.
- β_c . Higher average degree lowers the epidemic threshold; diseases spread more easily in denser networks.
- **Growth exponent** α . Epidemics grow exponentially when $R_0 > 1$; speed depends on both transmission and network density.
- **Limitation.** This approach **cannot capture the effects of hubs or degree variability** — only the average matters. Two networks with the same $\langle k \rangle$ but very different degree distributions look identical here.

The next approach fixes exactly that.

Why "average isn't enough" — a concrete picture

Imagine two networks, each with 100 nodes and the same average degree $\langle k \rangle = 5$.

- **Network A** — a regular ring lattice: every node has *exactly* 5 contacts.
- **Network B** — one **hub** with 100 contacts, and 99 leaves each with ~ 1 contact. The average is still ~ 2 (close enough to make the point).

To Approach 1, both look identical — they have the same $\langle k \rangle$. But:

- An infection started in **Network A** spreads slowly and uniformly.
- An infection that reaches the **hub of Network B** explodes — the hub re-infects nearly everyone.

The mean-field calculation cannot distinguish the two cases. We need something better.

Approach 2 — Heterogeneous Mean-Field

Main idea. Account for differences in individual connectivity by **grouping nodes by degree class**.

All degree- k nodes share equations, but degree-2 nodes obey different equations from degree-100 nodes.

Assumption. Individuals with the same degree behave identically, but nodes can have different degrees.

- Captures the effect of **hubs** and **degree variability**
- Much more realistic for heterogeneous (e.g. scale-free) networks
- Mathematics is still ODE-based and tractable

Pioneered by Pastor-Satorras & Vespignani (2001).

A scaffolding question — what is "the chance a random contact is infected"?

In NS22 we used I/N — the global infected fraction. On a network, this is **wrong**.

Why? When you pick a random *edge* in a graph and follow it, you are biased toward nodes with many edges. *High-degree nodes are over-represented at the ends of random edges* — simply because they have more ends.

This is the same phenomenon behind the **friendship paradox** ("your friends have more friends than you do"), which we will meet again in Part 4.

So the right object — let's call it $\Theta(t)$ — is *the probability that a random end of a random edge points to an infected node*. The next slide writes it down.

SIR equations — Heterogeneous Mean-Field

$$\frac{dS_k(t)}{dt} = -\beta k S_k(t) \Theta(t)$$

$$\frac{dI_k(t)}{dt} = \beta k S_k(t) \Theta(t) - \gamma I_k(t)$$

$$\frac{dR_k(t)}{dt} = \gamma I_k(t)$$

Where:

- $S_k(t)$, $I_k(t)$, $R_k(t)$ — numbers of susceptible, infected, recovered nodes **of degree k**
- k — degree (number of contacts) of the class
- $\Theta(t)$ — probability a random edge-end points to an infected node (defined next)

Each degree class gets its own SIR equations. They are coupled through the global $\Theta(t)$.

Calculating $\Theta(t)$ — and why it is not I/N

$$\Theta(t) = \frac{\sum_{k'} k' P(k') I_{k'}(t)}{\langle k \rangle N}$$

- k' — possible degrees in the network
- $P(k')$ — fraction of nodes with degree k'
- $I_{k'}(t)$ — number of infected nodes of degree k' at time t
- $\langle k \rangle$ — average degree

How to read it.

- $k' P(k')$ — probability that following a random edge lands on a node of degree k' (high-degree nodes have more incoming edges, so they are over-represented)
- Multiply by the infected fraction in that class, sum over all degrees
- Divide by the total number of edge-ends ($\langle k \rangle N$) to get a probability

Intuition. The formula gives **more weight to infected nodes with many connections** — exactly the bias that the global I/N misses.

Caveat. This formula assumes the network is **uncorrelated** — the degree of a node's neighbour is

Where the equations come from — step by step

1. **Degree classes.** Nodes are grouped by degree k — all degree- k nodes share equations.
2. **Contact structure.** A node with degree k has k contacts.
3. **Probability of infectious contact.** Each of those contacts is with an infected node with probability $\Theta(t)$ (not I/N).
4. **Expected infectious contacts.** $k \cdot \Theta(t)$ for a degree- k node.
5. **Force of infection.** $\beta k \Theta(t)$ for each susceptible of degree k .

The factor of k in front is the key new ingredient — high-degree nodes feel the epidemic much more strongly than low-degree nodes do.

Key quantities — Heterogeneous Mean-Field

- **Basic reproduction number:** $R_0 = \frac{\beta}{\gamma} \cdot \frac{\langle k^2 \rangle}{\langle k \rangle}$
- **Critical transmission rate:** $\beta_c = \frac{\gamma \langle k \rangle}{\langle k^2 \rangle}$
- **Early-stage growth exponent:** $\alpha = \beta \frac{\langle k^2 \rangle}{\langle k \rangle} - \gamma = \gamma(R_0 - 1)$

The new factor is $\langle k^2 \rangle / \langle k \rangle$ — the **second moment over the first**. This single ratio is what makes network heterogeneity show up in the epidemic threshold.

Why the factor $\langle k^2 \rangle / \langle k \rangle$ matters

- $\langle k^2 \rangle$ — average of the *square* of the degree (captures **variance** in connectivity)
- $\langle k \rangle$ — average degree

For a **homogeneous** network (everyone has roughly the same degree), $\langle k^2 \rangle \approx \langle k \rangle^2$, so $\langle k^2 \rangle / \langle k \rangle \approx \langle k \rangle$ — and we recover the homogeneous result.

For a **heterogeneous** network (some nodes have far more connections than others), $\langle k^2 \rangle$ is dominated by the rare high-degree nodes — and can grow much faster than $\langle k \rangle$.

Consequence. The threshold β_c is much *lower* in heterogeneous networks — even weakly transmissible diseases can spread.

Tiny worked example. Take a 5-node network with degrees $\{1, 1, 1, 1, 16\}$ — four leaves and one big hub.

- $\langle k \rangle = 4$
- $\langle k^2 \rangle = (1 + 1 + 1 + 1 + 256)/5 = 52$
- $\langle k^2 \rangle / \langle k \rangle = 13$

So the *effective* degree that drives the epidemic is **13**, not the average **4**. The hub matters far more than its single contribution to the average suggests.

Worked example — vanishing threshold on scale-free networks

For a power-law degree distribution $P(k) \propto k^{-\gamma_{\text{net}}}$ (with exponent γ_{net} , not to be confused with the recovery rate γ):

Exponent γ_{net}	$\langle k^2 \rangle$ as $N \rightarrow \infty$	$\beta_c = \gamma \langle k \rangle / \langle k^2 \rangle$
$\gamma_{\text{net}} > 3$	finite	finite — ordinary threshold
$2 < \gamma_{\text{net}} \leq 3$	diverges with N	$\beta_c \rightarrow 0$ as $N \rightarrow \infty$

Many real networks (the Internet, sexual-contact networks, scientific collaboration) have γ_{net} in the range 2 – 3. So the famous Pastor-Satorras & Vespignani (2001) result:

On a large scale-free network with $\gamma_{\text{net}} \leq 3$, **the epidemic threshold effectively vanishes** — *any* contagious disease can spread.

This is the single most important result in network epidemiology.

Interpretation and consequences — Heterogeneous Mean-Field

- R_0 . Strongly affected by degree variance. Networks with hubs (large $\langle k^2 \rangle$) have much higher R_0 than the homogeneous model would predict.
- β_c . Can become extremely small in networks with large degree variance (e.g. scale-free networks) — even weakly transmissible diseases can cause epidemics.
- **Growth exponent α** . Early-epidemic growth is much faster in heterogeneous networks because hubs amplify the early spread.
- **Practical consequence. Targeting hubs** for intervention is dramatically more effective than random strategies. Whether this is *practical* depends on whether we can find the hubs — see Part 4.

From averages to individuals — the next step

Heterogeneous mean-field still **groups** nodes by degree. All degree-7 nodes are interchangeable; all degree-100 nodes are interchangeable.

But two degree-100 hubs in different parts of the network may have very different epidemic impact:

- A hub *inside* a tight community amplifies a local outbreak
- A hub *between* two communities can carry an outbreak from one to the other

To capture this, we need to drop aggregation entirely and **track every node by name** — using the full adjacency matrix.

That is the third approach.

Approach 3 — Individual-Based SIR on networks

Main idea. Track the state of *each individual node* using the **exact** network structure — no aggregation by degree class.

Assumption. Each node can have a unique set of neighbours; we use the full adjacency matrix.

- Most detailed and realistic
- Captures **stochastic effects** and **local network structure** (clustering, communities)
- Most expensive — needs $O(|V|^2)$ structure in dense regimes

This is what real epidemic simulators (EpiModel, GLEAM, ...) actually do.

SIR equations — Individual-Based

$$\begin{aligned}\frac{dS_i(t)}{dt} &= -\beta S_i(t) \sum_j A_{ij} I_j(t) \\ \frac{dI_i(t)}{dt} &= \beta S_i(t) \sum_j A_{ij} I_j(t) - \gamma I_i(t) \\ \frac{dR_i(t)}{dt} &= \gamma I_i(t)\end{aligned}$$

Where:

- $S_i(t), I_i(t), R_i(t)$ — probability that node i is susceptible, infected, or recovered at time t
- A_{ij} — adjacency matrix entry (1 if i and j are connected, 0 otherwise)

Each node has its own ODE — coupled through the adjacency matrix.

Where the equations come from — step by step

1. **Node-level tracking.** Each node i has its own equations.
2. **Contact structure.** The sum $\sum_j A_{ij} I_j(t)$ counts the *infected neighbours* of node i .
3. **Force of infection.** $\beta \sum_j A_{ij} I_j(t)$ is the rate at which node i acquires infection.
4. **Stochasticity.** The same equations can be simulated as a discrete Markov process (each node flips state with the appropriate probability per time step).

No more degree classes; no more averaging. Just the actual graph and the actual disease parameters.

Why eigenvalues? — motivating the spectral threshold

The next slide gives R_0 for the individual-based model in terms of $\lambda_{\max}$, the **largest eigenvalue of the adjacency matrix**. Why does this number govern the epidemic?

Intuition. Early on, infection spreads by repeated multiplication by the adjacency matrix: an infected node at step t infects its neighbours at step $t + 1$, and so on. After many steps, the dominant growth direction is the **principal eigenvector of A** , and the rate of growth is set by $\lambda_{\max}$.

In words: $\lambda_{\max}$ measures *how many self-amplifying paths the network supports* — the bigger it is, the more easily a small initial seed can compound into a full outbreak.

For sparse, locally tree-like networks, $\lambda_{\max} \approx \langle k^2 \rangle / \langle k \rangle$ — recovering the heterogeneous-mean-field result. The spectral view generalises it.

Key quantities — Individual-Based

- **Basic reproduction number:** $R_0 = \frac{\beta}{\gamma} \lambda_{\max}$

where $\lambda_{\max}$ is the largest eigenvalue of the adjacency matrix A — a measure of how strongly connected and influential the network is *as a whole*.

- **Per-node basic reproduction number:** $R_0(i) = \frac{\beta}{\gamma} v_i$

where v_i is node i 's **eigenvector centrality** — how well-connected i is, weighted by how central its neighbours are (recall NS06).

- **Critical transmission rate:** $\beta_c = \frac{\gamma}{\lambda_{\max}}$
- **Early-stage growth exponent:** $\alpha = \beta \lambda_{\max} - \gamma = \gamma(R_0 - 1)$

The same threshold structure as the previous models — just with $\lambda_{\max}$ replacing the degree statistic.

Interpretation and consequences — Individual-Based

- R_0 . Determined by the most "central" or influential nodes and the overall network structure (not just by the degree distribution).
- β_c . Sensitive to the specific patterns of connectivity; **a single well-connected hub can lower the threshold for the whole network.**
- **Growth exponent α .** Epidemics grow fastest in regions of the network associated with the largest-eigenvector components — typically hubs or bridges.
- **Practical consequence.** This approach allows **precise identification of critical nodes** for intervention and captures clustering, community structure, and other real-world network features.

Comparing the three approaches

Model	Network structure used	R_0	Critical threshold β_c
Homogeneous MF	Average degree	$\frac{\beta \langle k \rangle}{\gamma}$	$\frac{\gamma}{\langle k \rangle}$
Heterogeneous MF	Degree distribution	$\frac{\beta \langle k^2 \rangle}{\gamma \langle k \rangle}$	$\frac{\gamma \langle k \rangle}{\langle k^2 \rangle}$
Individual-based	Full adjacency matrix	$\frac{\beta}{\gamma} \lambda_{\max}$	$\frac{\gamma}{\lambda_{\max}}$

Summary. As we move from homogeneous $\rightarrow$ heterogeneous $\rightarrow$ individual-based, we trade **mathematical simplicity** for **realism**:

- Homogeneous MF is fine if the network is roughly regular
- Heterogeneous MF is essential whenever the degree distribution is broad (most real networks)
- Individual-based is needed when *who is connected to whom* matters at the level of named individuals

From models to interventions — the practical payoff

Everything so far has been *descriptive*: how a disease spreads on different kinds of networks. Now comes the *prescriptive* part: **what should we do about it?**

Network structure changes the answer dramatically. In a homogeneous-mixing world, every individual is equally important to vaccinate. On a real network, **some individuals are dramatically more important than others** — and the right intervention exploits that fact.

The next four slides walk through the canonical intervention strategies, ending with the one that needs almost no information at all.

Why network structure changes intervention design

In a homogeneous-mixing world, every individual is equally important to vaccinate. On a real network, **some individuals are dramatically more important than others** — and the right intervention exploits that fact.

The four canonical strategies, in increasing order of cleverness:

- **Random vaccination** — pick people uniformly at random
- **Degree-based (targeted)** — vaccinate the highest-degree nodes
- **Betweenness-based** — vaccinate the nodes that bridge communities
- **Acquaintance immunization** — exploit the friendship paradox to find hubs without knowing the network

Each makes different assumptions about *what we know* about the network.

Random vs. targeted vaccination

Random vaccination — vaccinate a fraction f of the population chosen uniformly at random.

- Easy to implement; no information about the network needed
- On a scale-free network, **a huge fraction of the population must be vaccinated** to interrupt transmission, because random vaccination rarely hits the hubs

Degree-based (targeted) vaccination — vaccinate the highest-degree nodes first.

- **Dramatically more efficient** on heterogeneous networks
- On a scale-free network, vaccinating just a few percent of hubs can break up the giant connected component and stop the epidemic
- The epidemic threshold, viewed as a function of the vaccinated fraction, drops far faster under targeted vaccination

The mathematics is unambiguous: **targeting beats random**, often by orders of magnitude.

The hub-information problem

Targeted vaccination has a catch — **we need to know who the hubs are.**

In real public-health settings:

- We rarely have a clean map of the contact network
- Asking people to self-report their contacts is unreliable and slow
- Privacy concerns block the kind of mass surveillance that would expose hubs directly
- For some networks (e.g. sexual-contact networks), the hub structure is precisely what people are reluctant to disclose

So we need a strategy that **achieves near-targeted efficiency without knowing the network.** The friendship paradox provides one.

The friendship paradox

Statement. *On average, your friends have more friends than you do.*

This is not a quirk of bad luck — it is a **mathematical inevitability** in any network with degree variance. (Feld, 1991.)

Why it is true. When you pick a random *person*, you sample uniformly. When you pick a *friend* of a random person — i.e., follow a random edge — you sample with bias toward high-degree nodes (they have more edges, so they appear more often at edge-ends).

This is exactly the bias that drove the $\Theta(t)$ formula in Part 2.

Consequence. A randomly chosen friend of a randomly chosen person tends to have **higher degree than the random person did** — without needing any knowledge of the global network.

Tiny worked example. A network of 5 people: A, B, C, D, E. Friendships: $A-B$, $A-C$, $A-D$, $A-E$ (A is the hub; everyone else has just one friend).

- *Random person's degree:* $(4 + 1 + 1 + 1 + 1)/5 = 1.6$
- *Random friend's degree (sample by following an edge):* $(4 + 4 + 4 + 4 + 1 + 1 + 1 + 1)/8 = 2.5$

Your friends really do have more friends than you do.

Acquaintance immunization

Cohen, Havlin & ben-Avraham (2003). A startlingly simple intervention that exploits the friendship paradox:

Pick a random person. Vaccinate one of their (randomly chosen) acquaintances.

That's it.

Why it works.

- The "friend" you reach is biased toward being a hub (friendship paradox)
- So this procedure preferentially vaccinates high-degree nodes — *without* needing to know who they are
- Achieves **near-targeted efficiency** with only local information (you only need to ask each person to nominate a contact)

Why it matters in practice.

- Trivial to implement — no contact-network mapping required
- Works in domains where targeted vaccination is impossible (sexual-contact networks, anonymous online communities, undocumented populations)
- Robust to incomplete information

Summary — what NS22 + NS23 give you

- **NS22.** The compartmental framework — SI, SIS, SIR, SEIR, SIRS — with $R_0 = \beta/\gamma$ as the universal threshold. Powerful for analytical insight under homogeneous mixing.
- **NS23.** Three ways to lift the homogeneous-mixing assumption — homogeneous MF, heterogeneous MF, individual-based — each replacing R_0 's scaling factor with a richer summary of the network.
- **Central result.** On heterogeneous (e.g. scale-free) networks, β_c can effectively vanish — and **targeting hubs is dramatically more effective than random intervention**. The friendship paradox lets us do this targeting with only local information.

The same machinery is now standard not only in epidemiology but also in misinformation control, viral marketing, and infrastructure-failure analysis.

Q&A

